

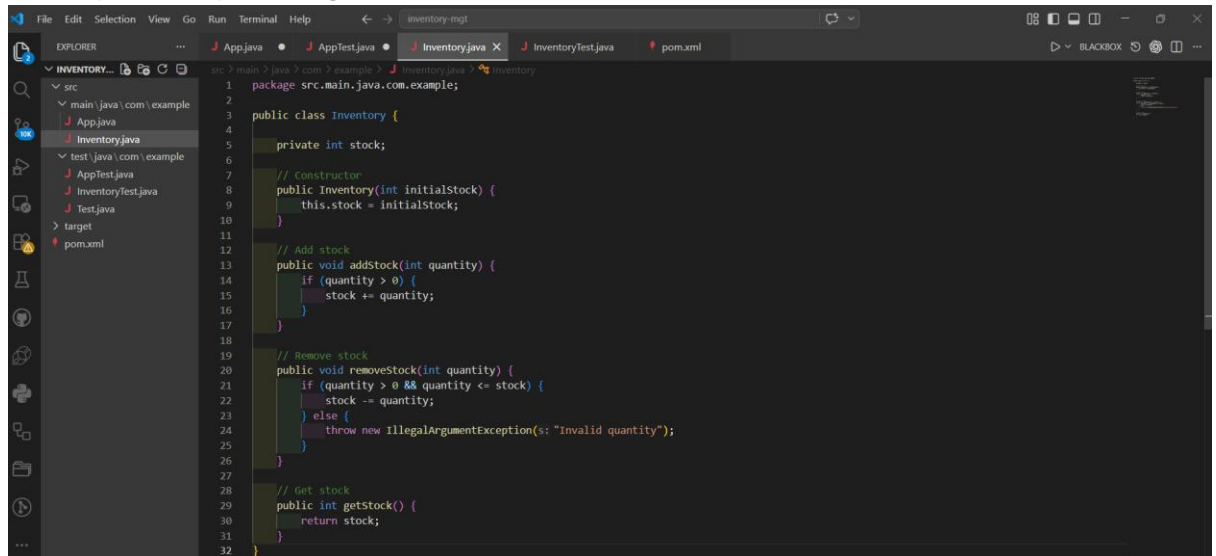
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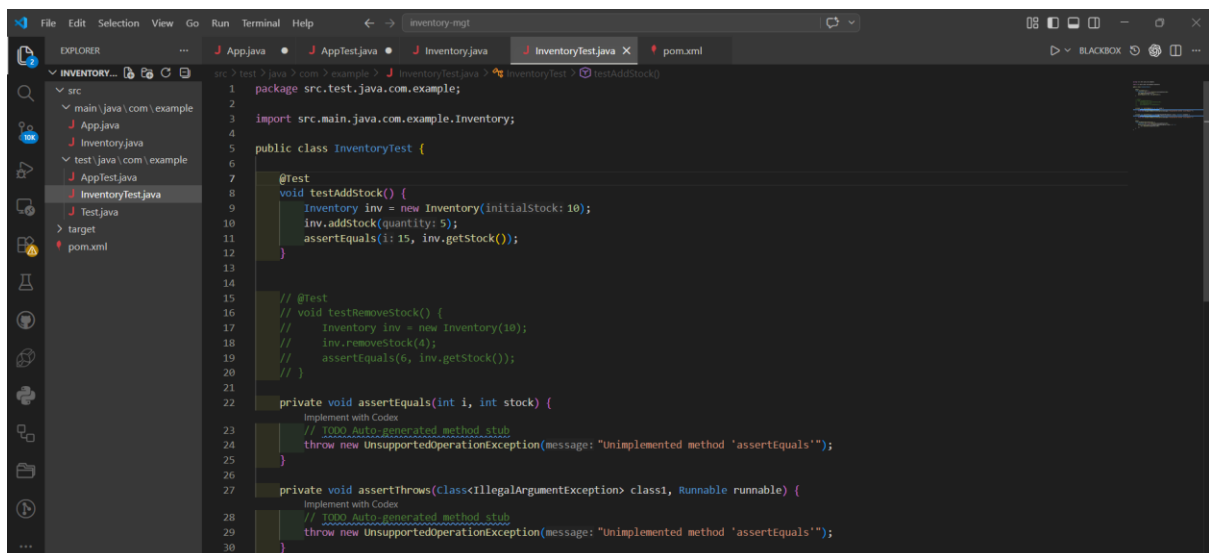
Course: Agile Development Process and DevOps

FAT LAB EXAM

1. Develop Stock Update Logic

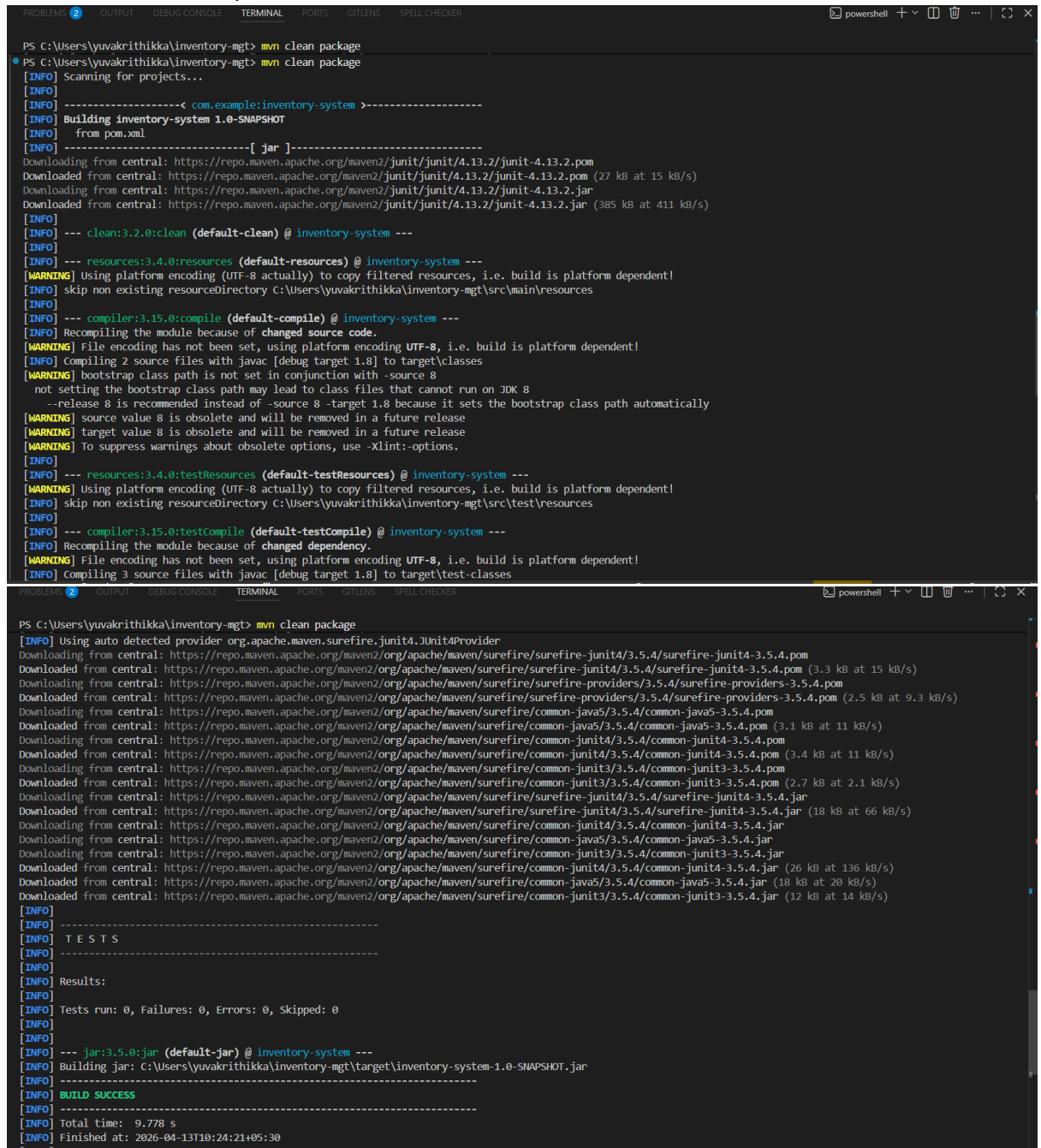


```
1 package src.main.java.com.example;
2
3 public class Inventory {
4     private int stock;
5
6     // Constructor
7     public Inventory(int initialStock) {
8         this.stock = initialStock;
9     }
10
11     // Add stock
12     public void addStock(int quantity) {
13         if (quantity > 0) {
14             stock += quantity;
15         }
16     }
17
18     // Remove stock
19     public void removeStock(int quantity) {
20         if (quantity > 0 && quantity <= stock) {
21             stock -= quantity;
22         } else {
23             throw new IllegalArgumentException(s: "Invalid quantity");
24         }
25     }
26
27     // Get stock
28     public int getStock() {
29         return stock;
30     }
31 }
32
```



```
1 package src.test.java.com.example;
2
3 import src.main.java.com.example.Inventory;
4
5 public class InventoryTest {
6
7     @test
8     void testAddStock() {
9         Inventory inv = new Inventory(initialStock: 10);
10         inv.addStock(quantity: 5);
11         assertEquals(i: 15, inv.getStock());
12     }
13
14     // @Test
15     void testRemoveStock() {
16         // Inventory inv = new Inventory(10);
17         // inv.removeStock(4);
18         // assertEquals(6, inv.getStock());
19     }
20
21     private void assertEquals(int i, int stock) {
22         // Implement with Codex
23         // 1000 Auto-generated method stub
24         throw new UnsupportedOperationException(message: "Unimplemented method 'assertEquals'");
25     }
26
27     private void assertThrows(Class<IllegalArgumentException> class1, Runnable runnable) {
28         // Implement with Codex
29         // 1000 Auto-generated method stub
30         throw new UnsupportedOperationException(message: "Unimplemented method 'assertThrows'");
31     }
32 }
```

2. Junit tests for stock updation:



```
PS C:\Users\yuvakrithikka\inventory-mgt> mvn clean package
PS C:\Users\yuvakrithikka\inventory-mgt> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:inventory-system >-----
[INFO] Building inventory-system 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/junit/junit/4.13.2/junit-4.13.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/4.13.2/junit-4.13.2.pom (27 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/junit/junit/4.13.2/junit-4.13.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/junit/junit/4.13.2/junit-4.13.2.jar (385 kB at 411 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ inventory-system ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ inventory-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\yuvakrithikka\inventory-mgt\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ inventory-system ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 2 source files with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ inventory-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\yuvakrithikka\inventory-mgt\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ inventory-system ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 3 source files with javac [debug target 1.8] to target\test-classes

PS C:\Users\yuvakrithikka\inventory-mgt> mvn clean package
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.pom (3.3 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.5.4/surefire-providers-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.5.4/surefire-providers-3.5.4.pom (2.5 kB at 9.3 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.5.4/common-java5-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.5.4/common-java5-3.5.4.pom (3.1 kB at 11 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.pom (3.4 kB at 11 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.pom (2.7 kB at 2.1 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit4/3.5.4/surefire-junit4-3.5.4.jar (18 kB at 66 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit4/3.5.4/common-junit4-3.5.4.jar (26 kB at 136 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.5.4/common-java5-3.5.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.5.4/common-java5-3.5.4.jar (18 kB at 20 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-junit3/3.5.4/common-junit3-3.5.4.jar (12 kB at 14 kB/s)
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Results:
[INFO]
[INFO] Tests run: 0, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ inventory-system ---
[INFO] Building jar: C:\Users\yuvakrithikka\inventory-mgt\target\inventory-system-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 9.778 s
[INFO] Finished at: 2026-04-13T10:24:21+05:30
[INFO]
```

3. CI/CD pipeline with automated testing

Jenkins / inventory-pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Script ?

```
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('clone') {
7       steps {
8         git branch: 'main', url: 'https://github.com/naveen252006/Inventory-Management.git'
9       }
10    }
11
12    stage('Build') {
13      steps {
14        bat 'mvn clean compile'
15      }
16    }
17  }
18 }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Advanced

Advanced ▾

Save Apply

Pipeline Script:

```
pipeline {
  agent any

  stages {

    stage('Clone') {
      steps {
        git branch: 'main', url: 'https://github.com/naveen252006/Inventory-Management.git'
      }
    }

    stage('Build') {
      steps {
        bat 'mvn clean compile'
      }
    }

    stage('Test') {
      steps {
```

```

        bat 'mvn test'
    }
}
}

post{
    success{
        echo 'Build Successful'
    }
    failure {
        echo 'Build Failed'
    }
}
}

```

4. Console output:

```

#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 158B 0.0s done
#1 DONE 0.1s

#2 [internal] load metadata for docker.io/library/node:18
#2 ...

#3 [auth] library/node:pull token for registry-1.docker.io
#3 DONE 0.0s

#2 [internal] load metadata for docker.io/library/node:18
#2 DONE 4.7s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.1s

#5 [internal] load build context
#5 transferring context: 2.38MB 1.9s done
#5 DONE 2.0s

#6 [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
#6 resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783 0.1s done
#6 sha256:461677a72fb7fe40d34a37d6a1958c4d16772d8dd77f572ec56a1f6c41a375d4 446B / 446B 0.7s done

```

```

#7 [2/5] WORKDIR /app
#7 DONE 1.4s

#8 [3/5] COPY package.json .
#8 DONE 0.1s

#9 [4/5] RUN npm install
#9 6.489
#9 6.489 added 68 packages, and audited 69 packages in 6s
#9 6.489
#9 6.489 15 packages are looking for funding
#9 6.489 run "npm fund" for details
#9 6.492
#9 6.492 found 0 vulnerabilities
#9 6.497 npm notice
#9 6.497 npm notice New major version of npm available! 10.8.2 -> 11.12.1
#9 6.497 npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.12.1
#9 6.497 npm notice To update run: npm install -g npm@11.12.1
#9 6.497 npm notice
#9 DONE 6.7s

#10 [5/5] COPY . .
#10 DONE 0.8s

```

```

#9 DONE 6.7s

#10 [5/5] COPY . .
#10 DONE 0.8s

#11 exporting to image
#11 exporting layers
#11 exporting layers 1.0s done
#11 exporting manifest sha256:47b769c454b02e508affc5eed06f968b54b9d04e1659aba95db8fb20972d8ccf 0.0s done
#11 exporting config sha256:db98e9c2d3373860fa5ce697d65d2d8437badc4c17d64acd4ed5f9fb0b492c8b 0.0s done
#11 exporting attestation manifest sha256:2633f28052084a016c30f1077a933b9fd701612e20c6ce05a596f47b34da1dcc 0.1s done
#11 exporting manifest list sha256:098df8eb217aa19445128339dc6be329cc2279e4aa5fadbda216bcd525c80cb9 0.0s done
#11 naming to docker.io/library/goal-tracker:latest
#11 naming to docker.io/library/goal-tracker:latest 0.0s done
#11 unpacking to docker.io/library/goal-tracker:latest
#11 unpacking to docker.io/library/goal-tracker:latest 2.3s done
#11 DONE 3.6s
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

5. Docker and Kubernetes:

The screenshot shows an IDE interface with the following components:

- EXPLORER:** A file tree on the left showing the project structure:
 - INVENTORY-MGT
 - src
 - main (java.com) example
 - App.java
 - Inventory.java
 - test (java.com) example
 - AppTest.java
 - InventoryTest.java
 - Test.java
 - target
 - deployment.yaml
 - Dockerfile
 - pom.xml
- deployment.yaml:** A YAML file defining a Kubernetes Deployment:


```

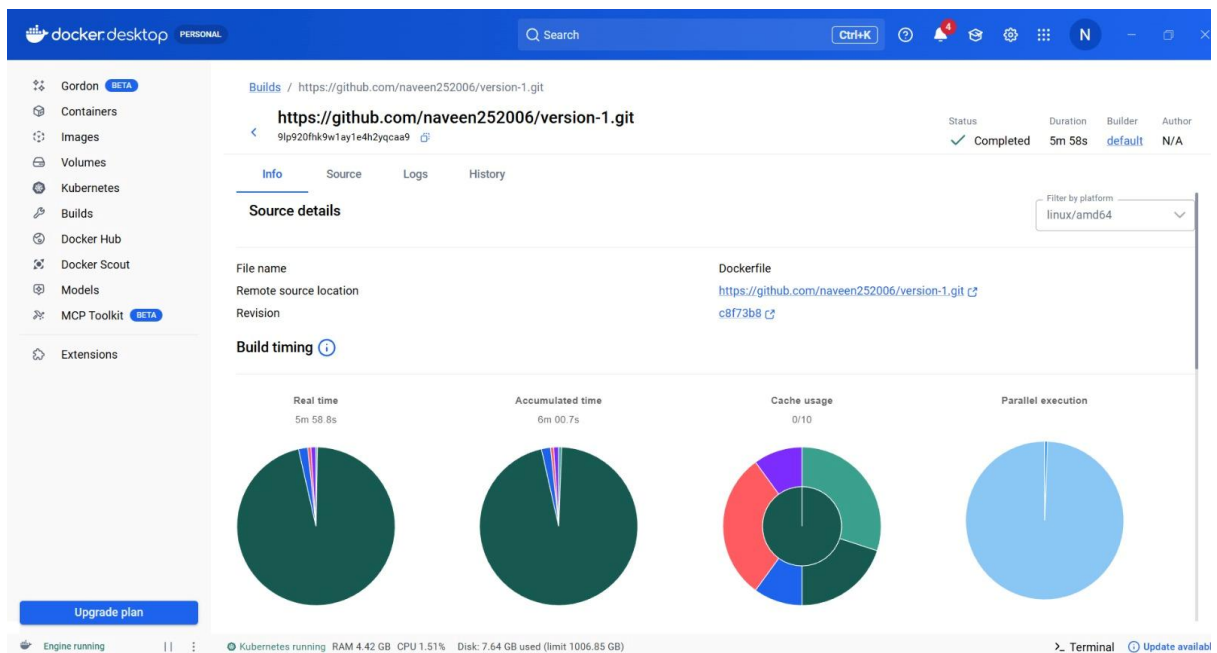
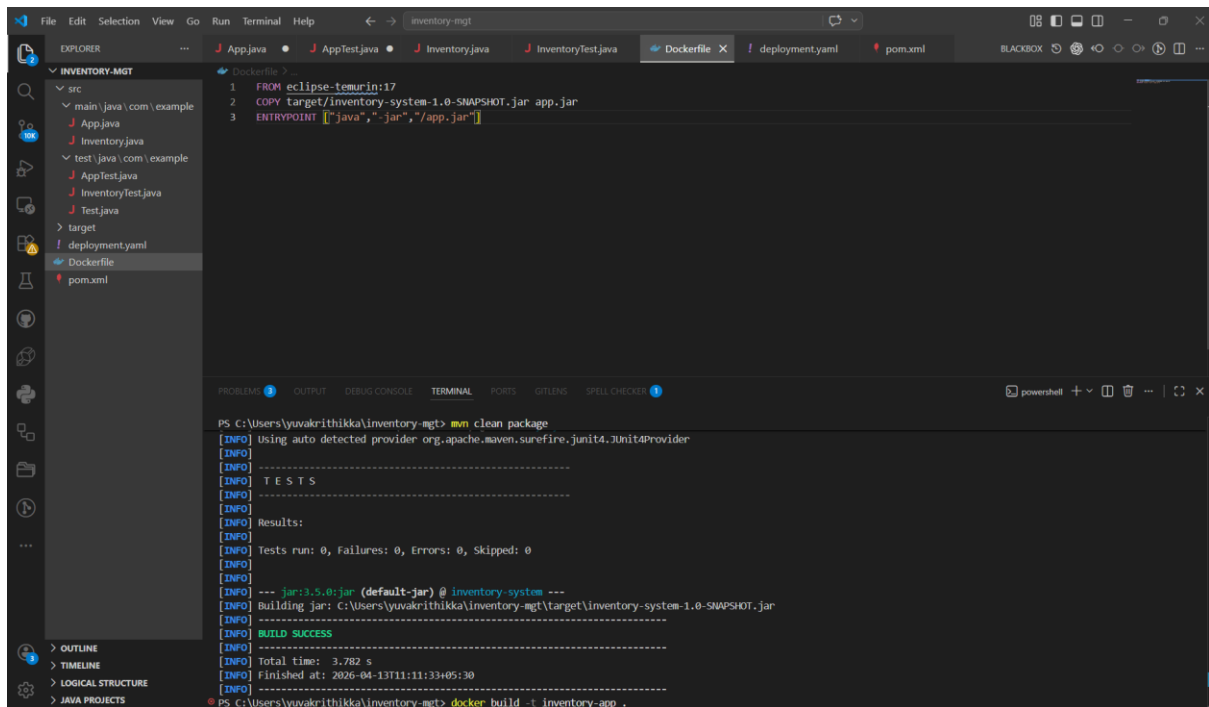
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: inventory-deployment
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: inventory
10  template:
11    metadata:
12      labels:
13        app: inventory
14    spec:
15      containers:
16      - name: inventory-container
17        image: username/inventory-app
18        ports:
19        - containerPort: 8080

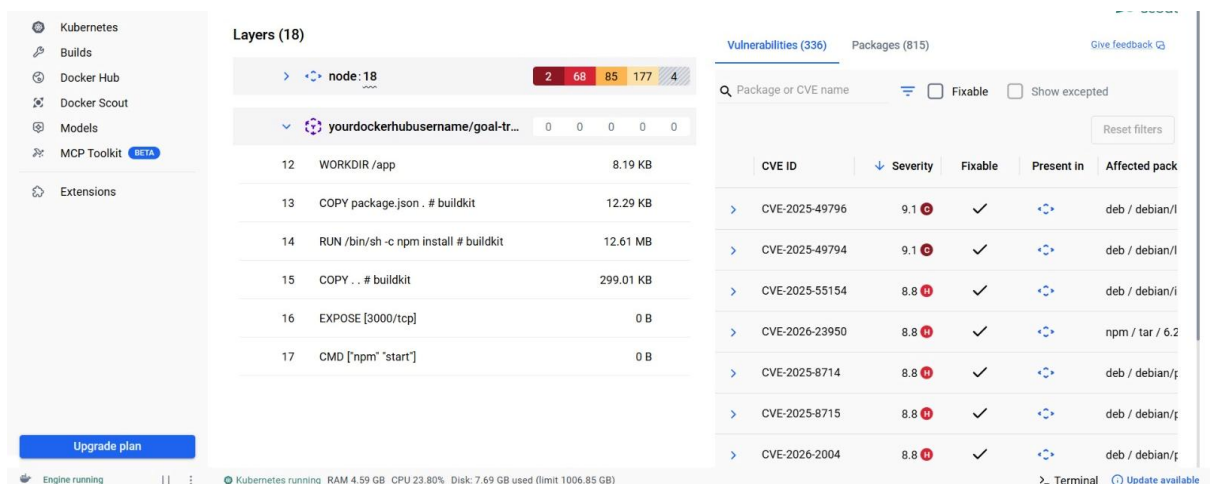
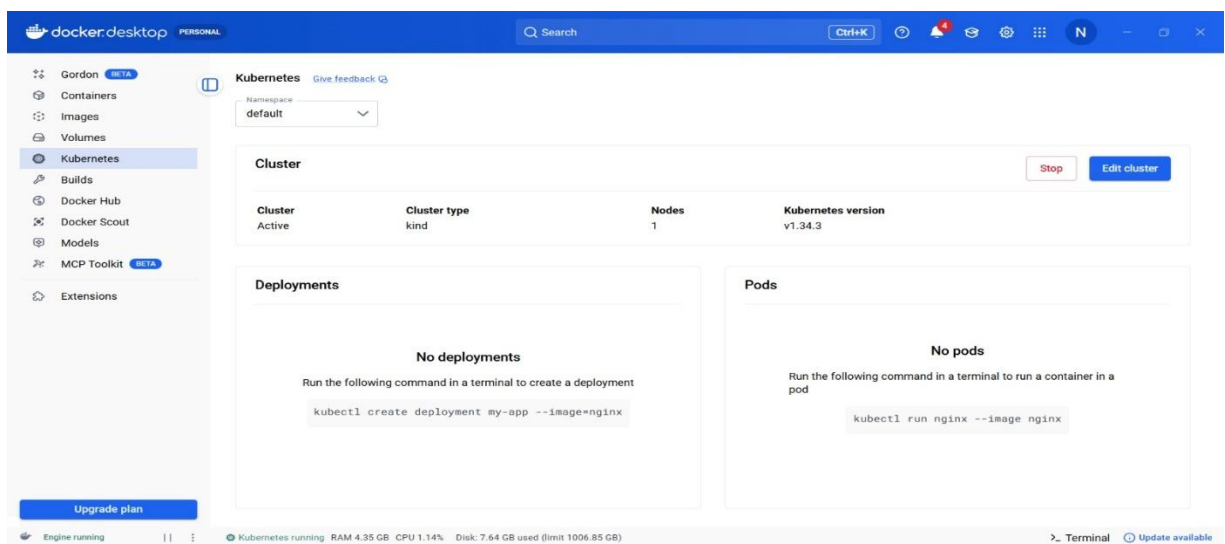
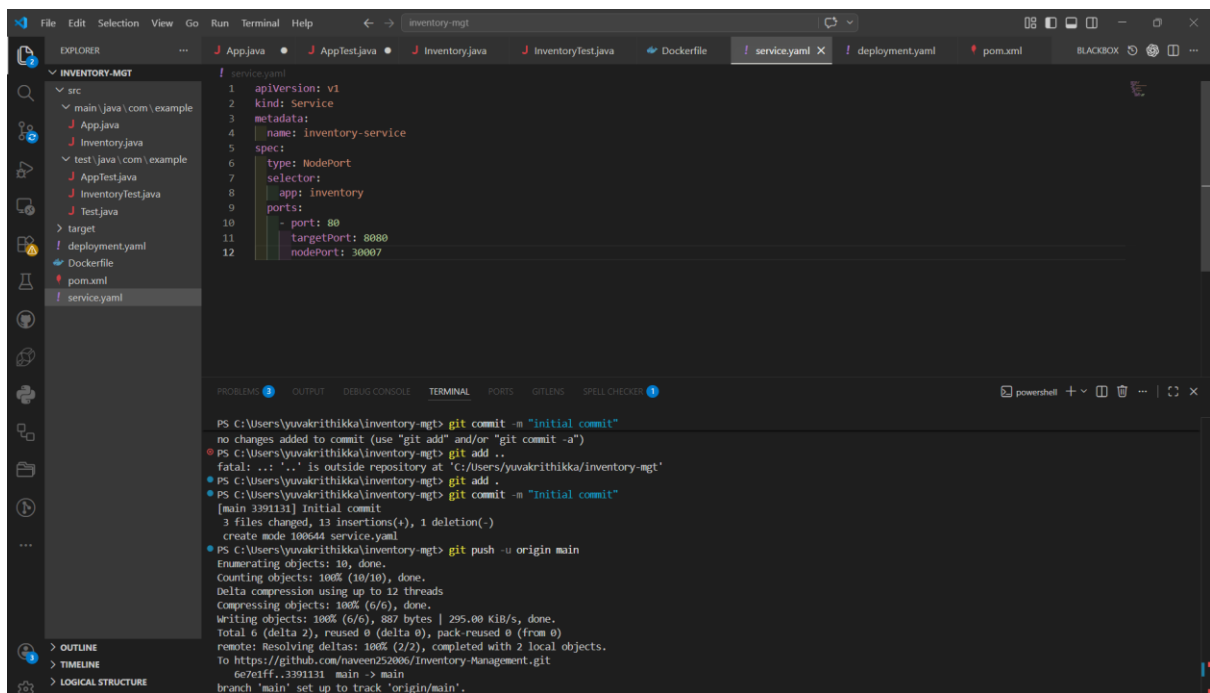
```
- TERMINAL:** A terminal window at the bottom showing the execution of Maven commands:


```

PS C:\Users\yuvakrithikka\inventory-mgt> mvn clean package
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 0, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ inventory-system ---
[INFO] Building jar: C:\Users\yuvakrithikka\inventory-mgt\target\inventory-system-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.782 s
[INFO] Finished at: 2026-04-13T11:11:33+05:30
[INFO]

```





Volumes

Kubernetes

Builds

Docker Hub

Docker Scout

Models

MCP Toolkit BETA

Extensions

Upgrade plan

13	COPY package.json . # buildkit	12.29 KB
14	RUN /bin/sh -c npm install # buildkit	12.61 MB
15	COPY ... # buildkit	299.01 KB
16	EXPOSE [3000/tcp]	0 B
17	CMD ["npm" "start"]	0 B

>	CVE-2025-49796	9.1 	✓	↔	deb / debian/l
>	CVE-2025-49794	9.1 	✓	↔	deb / debian/l
>	CVE-2025-55154	8.8 	✓	↔	deb / debian/i
>	CVE-2026-23950	8.8 	✓	↔	npm / tar / 6.2
>	CVE-2025-8714	8.8 	✓	↔	deb / debian/t
>	CVE-2025-8715	8.8 	✓	↔	deb / debian/t
>	CVE-2026-2004	8.8 	✓	↔	deb / debian/t
>	CVE-2026-2005	8.8 	✓	↔	deb / debian/t
>	CVE-2026-2006	8.8 	✓	↔	deb / debian/t
>	CVE-2025-9900	8.8 	✓	↔	deb / debian/t

1-10 of 336 | < > >>